

The 'Hyper-Fast' HYPERLOOP TRANSPORT SYSTEM



Deepak Halan
is associate
professor at School
of Management
Sciences, Apeejay
Styra University

Imagine waking up early morning in Mumbai, leaving for Pune to have breakfast with a friend there, and returning that same morning to attend your Mumbai office in time. Yes, it is possible with the Hyperloop Transportation System (HTS), which travels at supersonic speed of 1216 kilometres per hour, enabling you to cover the distance from Mumbai to Pune in less than twenty minutes. And all this without the routine long journey to the airport and the waiting in queues.

The HTS consists of high-pressure capsules moving in partially evacuated tubes at speeds of up to 1216kmph. This mode of transport is intended for super-fast travel between large cities and is the brainchild of Elon Musk, general director, Tesla Motors, who is also the creator of payment system PayPal and the SpaceX company, the first non-governmental organisation that successfully put a spaceship in orbit.

The Hyperloop system works on renewable energy and promises to transport people from one point to another in little over half the average time taken today by airplanes. The cost works out to about less than half the air-ticket price and hence the system can prove to be the right solution for high-traffic city pairs that are less than 1500km apart.

Comparatively, it is safer, faster, cheaper, more convenient, immune to

weather, environment-friendly and sustainably self-powered. It will bring radical change to the way we travel.

How Hyperloop works

In Hyperloop transportation, custom-designed capsules or pods zip smoothly through continuous steel tubes that are held at partial vacuum. The pod that sandwiches the passenger compartment between an air compressor upfront and a battery compartment in the rear is supported by air caster skis placed at the bottom. The skis float on a thin layer of air provided under high pressure, eliminating rolling resistance and allowing pod movement at high speeds. These capsules are planned to be driverless with speeds of more than 1000kmph. Linear induction motors placed along the tube control the pod speed. Electronically-assisted acceleration and braking determines the capsule speed.

During the pod's journey, an inlet fan and compressor push high-pressure air from the nose to tail. This action along with the partial vacuum eliminates most of the drag and boosts the speed. Low power consumption and reliance on existing infrastructure after re-engineering are the big positives. The complete system operation is designed to be automatic using extensive safety control systems with redundancy and supported by captive software.

The first Hyperloop transportation system in the world is under implementation between Los Angeles and San Francisco. It will travel this distance in flat 35 minutes with a planned capacity of 840 passengers per hour. The system will transport as many as six million passengers between Los Angeles and San Francisco every year.

Increase in the demand generated due to the lower cost of travelling on Hyperloop is expected to be met by shortening the time interval between capsule departures. Due to the short travel time and frequent

*An artist's rendition
of passengers riding
a Hyperloop capsule
(Source: <https://bit.ly/blogs.nytimes.com>)*

